Mirabilis-Tether: Formally Verified Cellular Hotspot Control via Pontryagin's Maximum Principle, Inverse-Gamma Fat-Tail Debiasing, and Clock-Offset-Cancelled One-Way Delay

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November 2026 — Version 1.0.0

Abstract

When a desktop or laptop operating system tethers to a smartphone Personal Hotspot (5G/LTE over Wi-Fi or USB), six interacting systems pathologies degrade interactive latency: (1) multi-megabyte host TCP socket buffers saturate un-managed smartphone baseband queues, inducing 450–1,200+ ms bufferbloat stalls; (2) symmetric ICMP Echo RTT controllers conflate uplink and downlink congestion, falsely throttling download capacity during heavy uploads; (3) fixed-step multiplicative-decrease (AIMD) controllers require 5–6 seconds to track abrupt cellular fades; (4) naive Gaussian or plug-in Pareto estimators underestimate heavy-tailed 5G/LTE scheduling spikes due to finite-sample Maximum Likelihood bias; (5) periodic Apple Wireless Direct Link (`awdl0`) social-channel scans induce 120 ms Wi-Fi radio stutters; and (6) cascading GTP-U, RFC 7335 CLAT46 (`464XLAT`), and overlay VPN (`utun*`) headers trigger silent Path MTU blackholes.

We present **Mirabilis-Tether** (`tether-optimize`), a zero-dependency macOS/XNU kernel optimization engine and native menu bar suite whose entire mathematical and state-transition specification is machine-checked in **110 zero-axiom constructive Lean 4 theorems** (`TetherOptimizeSpec.lean`, 4,099 lines, zero external imports) and validated at runtime by **72 live system assertions**. Our mathematical framework unifies: (i) *Mirabilis Abbreviatio*, an exact small-sample debiasing of the Hill–Taleb Pareto tail estimator $\widehat{\alpha}_{\mathrm{MLE}} \sim \mathrm{InvGamma}(n, n\alpha)$ paired with the Cirillo–Taleb compact-support Shadow Mean; (ii) a discrete formulation of **Pontryagin's Maximum Principle** over logarithmic Kelly utility solved in $O(1)$ integer arithmetic via the floor-division Galois connection; (iii) a **Revealed-Capacity Costate Dual** driven by 64-bit `NET_RT_IFLIST2` kernel counters that captures abrupt $100 \to 12$ Mbps cellular fades in a single 500 ms step ($11\times$ faster than AIMD); and (iv) a parallel 3-reflector BSD `IPPROTO_ICMP` **RFC 792 Type 13/14 One-Way Delay quorum** that algebraically cancels arbitrary host-reflector clock offsets $\theta \in \mathbb{Z}$ while rejecting single-reflector spikes via Birkhoff's distributive median. Live evaluation on Apple Silicon (M1) demonstrates a $697\times$ counter-sensing speedup (7.8 μs vs. 5.41 ms), 0.0% standby CPU overhead, exact wire-level θ-cancellation, and zero packet fragmentation or retransmissions.

1 Introduction

Modern cellular networks (5G NR and LTE-Advanced) exhibit extreme non-stationarity: air-interface capacity fluctuates by an order of magnitude within hundreds of milliseconds due to carrier aggregation switches, millimeter-wave/mid-band handovers, RLC retransmissions, and physical shadowing. When a macOS host tethers to an iPhone, iPad, or Android device via Wi-Fi (`en0`) or USB (`en7`/`en17`), the host kernel treats the link as a stationary high-bandwidth Ethernet or WLAN interface.

Specifically, six structural mismatches arise between the host networking stack and the mobile hotspot:

1. **Baseband Bufferbloat**: Stock macOS permits TCP socket buffers to auto-scale up to 16 MiB. On an 8 Mbps (1,000 B/ms) cellular uplink, a single bulk flow can enqueue 16,777,216 B into the smartphone's baseband FIFO, creating up to 16.7 s of queueing delay behind which interactive DNS, SSH, and WebRTC packets stall.

2. **Symmetric RTT Blindness**: Existing user-space autorate tools probe the link with symmetric ICMP Echo (`Type 8/0`) requests. Because RTT $= d_{\uparrow} + d_{\downarrow}$, a pure upload saturation inflates RTT and causes symmetric controllers to slash the *download* shaper pipe by 50–80%, starving downlink throughput even when the cellular downlink is completely idle.

3. **Multi-Second Lag on Abrupt Fades**: Classical AIMD autorate loops decay the shaper ceiling multiplicatively ($r_{k+1} = \lfloor 0.8 r_k \rfloor$). If a user enters an elevator and cellular capacity drops from 100 Mbps to 12 Mbps, multiplicative decay requires 11 consecutive bloated ticks (5.5 s at 500 ms/tick) to converge, during which the baseband buffer overflows continuously.

4. **Fat-Tailed Scheduling Spikes and Small-Sample MLE Bias**: Cellular MAC scheduling delays follow power-law Pareto/Lomax distributions $P(X > x) = (x/L)^{-\alpha}$ rather than Gaussian tails. In small rolling windows ($n \in [4, 20]$), the Maximum Likelihood (Hill) estimator $\widehat{\alpha}_{\mathrm{MLE}} = n / \sum_{i=1}^{n} \ln(X_i/L)$ is Inverse-Gamma distributed with expectation $\mathbb{E}[\widehat{\alpha}_{\mathrm{MLE}}] = \frac{n}{n-1}\alpha > \alpha$, systematically overestimating tail thinness and underestimating spike risk.

5. **AWDL Social-Channel Radio Stutters**: On macOS, the Apple Wireless Direct Link (`awdl0`) interface periodically hops off the active Wi-Fi AP channel to scan social channels (AirDrop/Handoff), introducing periodic 80–150 ms latency spikes on Wi-Fi hotspots.

6. **Cascading Encapsulation Blackholes**: Carrier GTP-U tunnels and RFC 7335 CLAT46 (`464XLAT`) IPv4-over-IPv6 translation reduce effective path MTU to 1420–1450 B. When combined with overlay VPNs (Tailscale/WireGuard `utun*` adding 80 B), un-scrubbed TCP segments suffer silent fragmentation drops.

Contributions. In this paper, we describe the mathematical theory, XNU systems implementation, and formal verification of **Mirabilis-Tether** (`tether-optimize` v1.0.0):

- **Constructive Formal Verification (Section 2)**: We formalize the complete specification in pure constructive Lean 4 (`TetherOptimizeSpec.lean`) with **zero external imports and zero axioms** (`propext`, `Classical.choice`, and `Quot.sound` are completely absent), proving 110 machine-checked theorems.

- ***Mirabilis Abbreviatio* & Inverse-Gamma Debiasing (Section 3)**: Following Taleb's statistical program on fat tails, we prove the exact small-sample bias identity of the Inverse-Gamma Hill MLE, derive the Minimum Variance Unbiased Estimator $\widehat{\alpha}_0 = \frac{n-1}{S_n}$, and integrate the Cirillo–Taleb compact-support Shadow Mean into the optimal control loop.

- **Pontryagin Hamiltonian Control via Galois Duality (Section 4)**: We formulate rate adaptation as a discrete Pontryagin optimal control problem over Kelly logarithmic utility and prove via the floor-division Galois connection that $u^{\star}(\lambda) = \mathrm{clamp}_{[u_{\min}, u_{\max}]}(\lfloor W/\lambda \rfloor)$ globally maximizes the Hamiltonian $\mathcal{H}(q, u, -\lambda)$ over all integer rates.

- **Clock-Offset-Cancelled One-Way Delay & Revealed-Capacity Duality (Section 5)**: We prove that RFC 792 Type 13/14 timestamps cancel arbitrary clock offsets $\theta \in \mathbb{Z}$, that

Birkhoff's distributive median3 provides 2-of-3 Byzantine glitch immunity, and that the dual costate $\lambda_{\text{cap}}(W, \hat{r}) = \lfloor W/(\hat{r}+1) \rfloor + 1$ captures arbitrary capacity fades in a single 500 ms step.

- **Empirical Evaluation on Apple Silicon (Section 6)**: We benchmark the live system on an Apple M1 host, demonstrating a 697× speedup in interface byte sensing, 0.0% idle CPU, live wire-level θ-cancellation, and +60.2% tail-risk capture on real TCP ACK streams.

2 Algebraic & Lattice Foundations of Network State Control

All proofs in `TetherOptimizeSpec.lean` are constructed from first principles over inductive Peano naturals and integers so that every theorem reduces purely in the Lean 4 kernel without axioms.

2.1 Bounded Join-Semilattice Catamorphism & Interface Partition

Let $(L, \sqcup, \bot, \top)$ be a bounded join-semilattice of network operating modes (ordered from passive Home LAN/Wi-Fi $\bot$ through constrained Wi-Fi hotspot to active USB Personal Hotspot $\top$). Interface classification partitions hardware ports into three mutually disjoint sets:

$$\mathcal{P}_{\text{hotspot}} \cap \mathcal{P}_{\text{protected}} = \emptyset, \qquad \mathcal{P}_{\text{overlay}} \cap \mathcal{P}_{\text{hotspot}} = \emptyset,$$

where $\mathcal{P}_{\text{hotspot}} = \{\texttt{en0}_{\text{tether}}, \texttt{en7}, \texttt{en17}\}$ covers iOS ('172.20.10.0/28'), Android ('192.168.43.0/24', '192.168.49.0/24'), and RFC 7335 CLAT46 ('192.0.0.0/29'), while $\mathcal{P}_{\text{protected}} = \{\texttt{en8}, \texttt{en9}, \texttt{en12}, \texttt{en13}, \texttt{en22}\}$ protects downstream Ethernet bridges and VMware USB passthrough NICs (**Theorems 1–12**).

Theorem 2.1 (Semilattice Catamorphism Uniqueness — Theorem 91)**.** *Let $f : A \to L$ map each active interface observation to a bounded join-semilattice $(L, \sqcup, \bot)$ with absorbing top $\top$. The fold catamorphism $\Phi_f([a_1, \ldots, a_k]) = f(a_1) \sqcup \cdots \sqcup f(a_k) \sqcup \bot$ is a monoid homomorphism under list concatenation, $\Phi_f(xs +\!\!+ ys) = \Phi_f(xs) \sqcup \Phi_f(ys)$, is idempotent on duplicate observations, and short-circuits identically upon encountering $\top$.*

2.2 Encapsulation Galois Adjunction & Dual PF Scrubbing

Consider a packet with inner TCP payload $p \in \mathbb{N}$ traversing an encapsulation stack of total header overhead $H \in \mathbb{N}$ over a link of Maximum Transmission Unit $M \in \mathbb{N}$. Define the monotone maps:

$$\text{encap}_H(p) = p + H, \qquad \text{decap}_H(M) = M \ominus H = \max(0, M - H).$$

Theorem 2.2 (Encapsulation Galois Connection — Theorem 93)**.** *For all $p, H, M \in \mathbb{N}$, $(\text{encap}_H, \text{decap}_H)$ forms a monotone Galois connection $\text{encap}_H \dashv \text{decap}_H$:*

$$\text{encap}_H(p) \le M \iff p \le \text{decap}_H(M).$$

Consequently, for compound encapsulation (inner TCP/IPv6 $H_1 = 60$ B inside WireGuard/Tailscale `utun*` $H_2 = 80$ B over cellular MTU $M = 1420$ B), the maximum non-fragmenting inner MSS is the composition of right adjoints:

$$\text{MSS}_{\text{inner}} \le \text{decap}_{H_1}(\text{decap}_{H_2}(M)) = M \ominus (H_1 + H_2) = 1420 - 140 = 1280 \text{ B},$$

*and clamping $\text{MSS}_{\text{outer}} = 1380$ B and $\text{MSS}_{\text{overlay}} = 1220$ B via idempotent PF `max-mss` scrubbing ($\min(p, c_1, c_2) = \min(p, \min(c_1, c_2))$, **Theorem 94**) strictly prevents L3 fragmentation across all IPv4, IPv6, GTP-U, and CLAT46 paths (**Theorems 13–28**).*

2.3 Jacobson–Karels Adaptive Deadband Invariant

To prevent chattering under baseline wireless jitter, let $v \in \mathbb{N}$ denote the RFC 6298 EWMA mean deviation (RTTVAR). We define the adaptive hysteresis thresholds:

$$\Theta_{\text{high}}(v) = 30 + \min(60, 2v), \qquad \Theta_{\text{low}}(v) = 12 + \min(20, \lfloor v/2 \rfloor).$$

Theorem 2.3 (Uniform Deadband Separation — Theorem 95)**.** *For all $v \in \mathbb{N}$, the deadband width satisfies $\Theta_{\text{high}}(v) - \Theta_{\text{low}}(v) \geq 18$ ms and is monotonically non-decreasing in v. Thus the bloated ($\Delta q > \Theta_{\text{high}}(v)$) and drained ($\Delta q \leq \Theta_{\text{low}}(v)$) decision regions are strictly separated for all $v \geq 0$.*

3 *Mirabilis Abbreviatio*: Inverse-Gamma Fat-Tail Estimation & Shadow Means

Cellular queueing spikes above a baseline threshold $L > 0$ follow a Pareto Type I tail:

$$f(x \mid \alpha, L) = \frac{\alpha L^{\alpha}}{x^{\alpha+1}}, \quad x \geq L, \quad \alpha > 0.$$

Given n exceedance observations $X_1, \ldots, X_n \geq L$, the sufficient statistic $S_n = \sum_{i=1}^{n} \ln(X_i/L)$ satisfies $S_n \sim \text{Gamma}(n, \alpha)$ with density $g(s) = \frac{\alpha^n}{\Gamma(n)} s^{n-1} e^{-\alpha s}$.

3.1 Exact Small-Sample Bias of the Hill MLE and MVUE Debiasing

The Maximum Likelihood (Hill) estimator is $\widehat{\alpha}_{\text{MLE}} = \frac{n}{S_n}$. Since $S_n \sim \text{Gamma}(n, \alpha)$, its reciprocal follows an Inverse-Gamma distribution:

$$\widehat{\alpha}_{\text{MLE}} = \frac{n}{S_n} \sim \text{InvGamma}(\text{shape} = n, \text{scale} = n\alpha).$$

For $n > 1$, the first moment of $\text{InvGamma}(n, \beta)$ is $\frac{\beta}{n-1}$, which yields:

$$\mathbb{E}[\widehat{\alpha}_{\text{MLE}}] = \frac{n\alpha}{n-1} = \left(1 + \frac{1}{n-1}\right)\alpha > \alpha.$$

In small real-time windows (e.g. $n = 4$), raw MLE overestimates α by $+33.3\%$, creating a dangerous illusion of thin tails (*Taleb's small-sample fat-tail trap*). Multiplying by the Bessel-like factor $\frac{n-1}{n}$ yields the exact Minimum Variance Unbiased Estimator (MVUE):

$$\widehat{\alpha}_0 = \frac{n-1}{S_n} = \frac{n-1}{n}\widehat{\alpha}_{\text{MLE}}, \qquad \mathbb{E}[\widehat{\alpha}_0] = \alpha \quad (\forall n \geq 2).$$

Theorem 3.1 (Inverse-Gamma Small-Sample Debiasing — Theorems 96 & 97)**.** *For all integer sample sizes $n \geq 2$ and positive log-exceedance sums $S_n > 0$, the debiased estimator $\widehat{\alpha}_0(n, S_n) = \lfloor (n-1)/S_n \rfloor$ satisfies $(n-1)\widehat{\alpha}_{\text{MLE}} = n\widehat{\alpha}_0$ in exact scaled arithmetic, strictly bounds $\widehat{\alpha}_0 < \widehat{\alpha}_{\text{MLE}}$, and converges monotonically toward $\widehat{\alpha}_{\text{MLE}}$ as $n \to \infty$.*

3.2 Posterior Predictive Lomax Tail & Cirillo–Taleb Shadow Mean

Integrating out the uncertain parameter α against the $\text{Gamma}(n, S_n)$ likelihood yields the exact Bayesian posterior predictive survival function (a Lomax / Pareto Type II distribution):

$$P(X_{\text{new}} > x \mid S_n, n) = \int_0^{\infty} \left(\frac{x}{L}\right)^{-\alpha} \frac{S_n^n}{\Gamma(n)} \alpha^{n-1} e^{-\alpha S_n} \, d\alpha = \left(1 + \frac{\ln(x/L)}{S_n}\right)^{-n},$$

with exact p-quantile $\mathrm{VaR}_p^\star = L\exp\Big(S_n((1-p)^{-1/n}-1)\Big)$ (**Theorem 99**).

When $\widehat{\alpha}_0 \le 1$ (the Cauchy/Lévy infinite-mean regime), the unconditional expectation $\mathbb{E}[X] = \infty$, but physical cellular timeouts truncate observations at $H = 350$ ms. Following Cirillo and Taleb, we compute the dual-bounded **Shadow Mean** on $[L, H]$:

$$\mu_{\text{shadow}}(\alpha, L, H) = \begin{cases} \dfrac{\alpha L}{\alpha - 1}\,\dfrac{1-(L/H)^{\alpha-1}}{1-(L/H)^{\alpha}}, & \alpha \neq 1, \\ \dfrac{L\ln(H/L)}{1-L/H}, & \alpha = 1, \end{cases}$$

which is strictly finite, continuous across $\alpha = 1$, and strictly exceeds the naive sample mean throughout the heavy-tailed regime $\alpha \in (0, 2)$ (**Theorem 98**).

4 Pontryagin's Maximum Principle & Galois Control Duality

Rather than tuning heuristic AIMD step sizes, we cast cellular rate control as a discrete-time optimal control problem via **Pontryagin's Maximum Principle (PMP)**.

4.1 The Discrete Kelly–Pontryagin Hamiltonian

Let $q_k \in \mathbb{N}$ be the baseband queue state, $u_k \in [u_{\min}, u_{\max}] \subset \mathbb{N}_+$ be the dummynet shaper control rate (in Mbps), $\tau_0 = 20$ ms be the target queue delay, and $W = u_{\max}\tau_0$ be the target Bandwidth-Delay Product scale. Define the discrete Kelly logarithmic utility:

$$U_W(u) = \sum_{j=1}^{u} \left\lfloor \frac{W}{j} \right\rfloor, \qquad \Delta U_W(u) = U_W(u) - U_W(u-1) = \left\lfloor \frac{W}{u} \right\rfloor.$$

With discrete queue dynamics $q_{k+1} = \max(0, q_k + u_k - c_k)$ and positive shadow price (costate) $\lambda_k \ge 1$ (in ms of marginal congestion cost per Mbps), the discrete Pontryagin Hamiltonian is:

$$\mathcal{H}(q_k, u, -\lambda_k) = U_W(u) + \lambda_k(u_{\max} - u).$$

Theorem 4.1 (Global Maximization of the Pontryagin Hamiltonian — Theorem 101)**.** *For every $u_{\min} \le u_{\max}$, $W \in \mathbb{N}$, and shadow price $\lambda \ge 1$, the closed-form integer control law*

$$u^\star(\lambda) = \mathrm{clamp}_{[u_{\min}, u_{\max}]}\left(\left\lfloor \frac{W}{\lambda} \right\rfloor\right)$$

globally maximizes $\mathcal{H}(q, u, -\lambda)$ over all $u \in [u_{\min}, u_{\max}]$:

$$\forall u \in [u_{\min}, u_{\max}], \qquad \mathcal{H}(q, u^\star(\lambda), -\lambda) \ge \mathcal{H}(q, u, -\lambda).$$

Proof Sketch (Formalized in Lean 4 lines 3130–3340). The forward difference of the Hamiltonian at step u is $\mathcal{H}(u) - \mathcal{H}(u-1) = \lfloor W/u \rfloor - \lambda$. By the fundamental Galois connection of integer floor division (`peanoDiv_galois`: $q \cdot d \le W \iff q \le \lfloor W/d \rfloor$) combined with commutativity of multiplication,

$$\left\lfloor \frac{W}{u} \right\rfloor \ge \lambda \iff \lambda \cdot u \le W \iff u \cdot \lambda \le W \iff u \le \left\lfloor \frac{W}{\lambda} \right\rfloor.$$

Thus $\mathcal{H}(q, u, -\lambda)$ is strictly non-decreasing for all $u \le \lfloor W/\lambda \rfloor$ and strictly decreasing for all $u > \lfloor W/\lambda \rfloor$. Clamping $\lfloor W/\lambda \rfloor$ to the convex interval $[u_{\min}, u_{\max}]$ therefore yields the exact global maximum on $[u_{\min}, u_{\max}]$. □

4.2 Taleb-Forced Adjoint Equation & Lyapunov Costate Relaxation

The costate λ_k is driven by the **Taleb-Forced Adjoint Target** (**Theorem 104**):

$$\Psi(\tau_0, \Delta q_k, \widehat{\alpha}_0, \mu_{\text{shadow}}) = \tau_0 + \Delta q_k + \min\left(120,\ \left\lfloor \frac{\max(0, 2000 - 1000\widehat{\alpha}_0)\,\mu_{\text{shadow}}}{2000} \right\rfloor\right),$$

which vanishes the tail risk premium when $\widehat{\alpha}_0 \geq 2$ (finite variance) and amplifies the shadow price proportionally to the Cirillo–Taleb Shadow Mean when $\widehat{\alpha}_0 < 2$. When the queue drains, λ_k relaxes toward τ_0 via $\lambda_{k+1} = \tau_0 + \lfloor 3(\lambda_k - \tau_0)/4 \rfloor$, which is a strict discrete Lyapunov contraction on $V(\lambda) = \lambda - \tau_0$ (**Theorem 105**).

5 Closing the Real-World Gap: OWD Quorum & Revealed-Capacity Duality

5.1 RFC 792 One-Way Delay Clock-Offset Cancellation

Let $\theta \in \mathbb{Z}$ be the unknown clock offset (plus any sub-day timezone discrepancy) between the macOS host's UTC clock and a remote ICMP reflector's UTC midnight clock. An RFC 792 Type 13/14 Timestamp exchange records four timestamps (T_1, T_2, T_3, T_4), producing apparent one-way delays:

$$\tilde{d}_\uparrow = T_2 - T_1 = b_\uparrow + \Delta q_\uparrow + \theta, \qquad \tilde{d}_\downarrow = T_4 - T_3 = b_\downarrow + \Delta q_\downarrow - \theta,$$

and apparent minimum baselines $\tilde{b}_\uparrow = b_\uparrow + \theta$, $\tilde{b}_\downarrow = b_\downarrow - \theta$.

Theorem 5.1 (Exact Clock-Offset Cancellation & Conservation — Theorem 106)**.** *For all $b_\uparrow, \Delta q_\uparrow, b_\downarrow, \Delta q_\downarrow \in \mathbb{N}$ and all clock offsets $\theta \in \mathbb{Z}$:*

$$\tilde{d}_\uparrow - \tilde{b}_\uparrow = \Delta q_\uparrow, \qquad \tilde{d}_\downarrow - \tilde{b}_\downarrow = \Delta q_\downarrow, \qquad \tilde{d}_\uparrow + \tilde{d}_\downarrow = (b_\uparrow + \Delta q_\uparrow) + (b_\downarrow + \Delta q_\downarrow) = \text{RTT}.$$

Moreover, whenever the link is in an empty-queue state (RTT $\leq$ baseRTT+4 ms), `tether-events` re-anchors $(\tilde{b}_\uparrow, \tilde{b}_\downarrow)$ so continuous quartz crystal clock skew ($\approx$ 10–50 ppm) never accumulates into false one-way bloat.

5.2 Byzantine 2-of-3 Multi-Reflector Quorum

To eliminate false throttling from single-reflector ICMP rate-limiting, `tether-events` probes three diverse anycast reflectors in parallel over non-privileged BSD `SOCK_DGRAM IPPROTO_ICMP` sockets (`9.9.9.9` Type 13, `149.112.112.112` Type 13, and `1.1.1.1` Type 8) and aggregates RTTs via Birkhoff's distributive lattice median:

$$\text{median3}(a, b, c) = \max(\min(a, b),\ \min(\max(a, b), c)).$$

Theorem 5.2 (Byzantine Single-Reflector Glitch Immunity — Theorem 107)**.** *If two reflectors report true delays $r_1, r_2 \in [L, H]$ and any one reflector suffers an arbitrary Byzantine spike $B \geq H$, then* median3 *lies strictly in $[L, H]$ across all $3! = 6$ permutations of (r_1, r_2, B).*

5.3 Revealed-Capacity Costate Dual: 1-Step Cellular Fade Capture

Suppose the shaper ceiling is $u_{\max} = 100$ Mbps ($W = u_{\max}\tau_0 = 2000$), and the user steps into an elevator where actual cellular delivery drops to $\hat{r} = 12$ Mbps (measured over 500 ms via 64-bit `NET_RT_IFLIST2` counters) while bloat triggers.

A naive inversion $\lambda_{\text{naive}} = \lfloor W/\hat{r} \rfloor$ fails because floor division rounds down, so $\lfloor W/\lfloor W/\hat{r} \rfloor \rfloor$ can exceed $\hat{r}$ (for $W = 100, \hat{r} = 30$, $\lfloor 100/30 \rfloor = 3 \implies \lfloor 100/3 \rfloor = 33 > 30$). Instead, we define the **Revealed-Capacity Costate Dual**:

$$\lambda_{\text{cap}}(W, \hat{r}) = \left\lfloor \frac{W}{\hat{r} + 1} \right\rfloor + 1.$$

Theorem 5.3 (Revealed-Capacity Galois Dual & 1-Step Fade Capture — Theorem 108)**.** *For all $W, \hat{r} \in \mathbb{N}$:*

1. $\lambda_{\text{cap}}(W, \hat{r}) \geq 1$ *(never divides by zero, even if $\hat{r} = 0$);*
2. $\left\lfloor \dfrac{W}{\lambda_{\text{cap}}(W, \hat{r})} \right\rfloor \leq \hat{r}$ *(the induced Pontryagin rate never exceeds revealed capacity $\hat{r}$);*
3. *For $W = 2000$ and an elevator fade $\hat{r} = 12$ Mbps, $\lambda_{\text{cap}}(2000, 12) = 154$ ms and $u^\star(154) = \lfloor 2000/154 \rfloor = 12$ Mbps in a **single 500 ms control step** ($11\times$ faster than AIMD).*

5.4 BDP-Isochronous CoDel/GRED Scaling & Two-Speed `kqueue` Cadence

Finally, to keep internal `dummynet` GRED marking thresholds isochronous in milliseconds across $r \in [2, 100]$ Mbps, we scale:

$$\text{min_th}(r) = \text{clamp}_{[2,24]}(\lfloor r/4 \rfloor + 2), \quad \text{max_th}(r) = \text{clamp}_{[6,64]}(\lfloor 3r/4 \rfloor + 6), \quad \text{slots}(r) = \text{clamp}_{[10,90]}(r + 10),$$

preserving $2 \leq \text{min_th}(r) < \text{max_th}(r) < \text{slots}(r) \leq 90$ for all r (**Theorem 109**), while `tether-events` switches its `kqueue` `EVFILT_TIMER` between 3000 ms in idle (Δbytes < 48 KB, emitting 0 radio probes) and 500 ms under active load (**Theorem 110**).

6 XNU Systems Implementation & Empirical Evaluation

We evaluated **Mirabilis-Tether** v1.0.0 on an Apple M1 Mac running macOS 26.5.2 (XNU kernel), Lean 4.35.0, and Wireshark 4.6 (`dumpcap`/`tshark`).

Table 1: Live Empirical Measurements on Apple Silicon (M1, macOS 26.5.2)

Subsystem / Metric	**Baseline / Legacy**	**Mirabilis-Tether v1.0.0**	**Measured Gain / Verificat**
64-bit Byte Counter Read	`netstat -ibn` (5.41 ms)	`sysctl(NET_RT_IFLIST2)`	**7.8** μs (**697**$\times$ speedup, 0 forks
Reactor Idle CPU & RSS	Shell loop (0.4% CPU)	`tether-events` (`kqueue`)	**0.0**% CPU, **2.64** MB RSS, 0 i
Menu Bar App Footprint	Electron tray ($>$ 150 MB)	`TetherMenubar.app` (Cocoa)	**0.0**% CPU, **13.0** MB RSS
RFC 792 OWD θ-Cancel	Raw $\tilde{d}_\uparrow = +1567$ ms	$\tilde{d}_\uparrow + \tilde{d}_\downarrow = 1567 - 1557$	**10** ms ($\theta \approx +1.56$ s cancelled;
4s Wire Audit (791 pkts)	Unchecked L3/L4 framing	`dumpcap -s 128` + Lua	Max L3 **1480** B ($\leq$ 1492), `fra`
ACK Tail Risk ($n = 19$)	Naive $\text{VaR}_{99} = 137.8$ ms	InvGamma $\widehat{\alpha}_0 = 1.22$	Shadow **21.4** ms, $\text{VaR}^\star_{99} =$ **22**
Formal Spec Verification	0 proofs	`TetherOptimizeSpec.lean`	**110**/**110** theorems in 17.5 s (**0**
Live System Self-Tests	Manual inspection	`tether-optimize test`	**72**/**72** assertions passed (0 fail

ARM64 Alignment-Safe Kernel ABI. On 64-bit ARM64 macOS, variable-length routing messages returned by `sysctl(NET_RT_IFLIST2)` can land on 4-byte boundaries (offset mod 8 = 4). Under `swiftc -O`, typed pointer dereferencing emits aligned 64-bit loads that fault or misread `if_msghdr2`; `tether-events` filters in-kernel via `mib[5] = Int32(if_nametoindex(iface))` and copies headers via alignment-safe `memcpy(&hdr, ...)`, achieving 7.8 μs per read.

7 Conclusion

By uniting constructive theorem proving in Lean 4, Taleb's small-sample fat-tail probability theory, Pontryagin's Maximum Principle, and low-level XNU kernel event primitives, **Mirabilis-Tether** transforms mobile tethering from brittle heuristic scripting into a formally verified, zero-maintenance optimal control system.